

# Scientific sampling event log from R/V Roger Revelle cruise RR2201 in the Eastern Indian Ocean (Argo Basin) from January to March 2022

**Website:** <https://www.bco-dmo.org/dataset/943418>

**Data Type:** Cruise Results

**Version:** 1

**Version Date:** 2024-11-13

## Project

» [Collaborative Research: Mesoscale variability in nitrogen sources and food-web dynamics supporting larval southern bluefin tuna in the eastern Indian Ocean](#) (BLOOFINZ-IO)

## Program

» [Second International Indian Ocean Expedition](#) (IIOE-2)

| Contributors                       | Affiliation                                           | Role                      |
|------------------------------------|-------------------------------------------------------|---------------------------|
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## Abstract

This event log contains unique event numbers, locations (latitude, longitude), dates, times, and related information for sampling operations conducted for the BLOOFINZ-IO Project on cruise RR2201 (R/V Roger Revelle) off of NW Australia in January-March 2021. Submitted data for this project will reference event numbers and sampling information in this file, and this file provides the broader context for related samplings that may have been done in temporal or spatial proximity to one another.

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## Coverage

**Location:** Northwest Australia, Argo Basin

**Spatial Extent:** N:9.3107 E:139.6872 S:-17.1854 W:113.9123

**Temporal Extent:** 2022-01-21 - 2022-03-11

## Methods & Sampling

Events were logged during the cruise using the R2R Eventlogger (<https://www.rvdata.us/about/event-log>).

The RR2201 cruise report can be found at <http://hdl.handle.net/1834/43464>.

## BCO-DMO Curation Notes

- Imported original file "Event Logger\_EXCELSpreadsheet\_ABBREVIATED\_7Nov2024xlsx.xlsx" into the BCO-DMO system.
- Marked 'nd' as a missing data value (missing data are empty/blank in the final csv file).
- Renamed fields to comply with BCO-DMO naming conventions.
- Converted the "DateTimeUTC" column to ISO 8601 format and renamed "ISO\_DateTime\_UTC".
- Saved final file as "943418\_v1\_rr2201\_event\_log.csv"

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## Data Files

| File                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------|
| <b>943418_v1_rr2201_event_log.csv</b> (Comma Separated Values (.csv), 220.27 KB)<br>MD5:00cb530ad6f5baf8b8b0d562c1927a78 |
| Primary data file for dataset ID 943418, version 1                                                                       |

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## Related Publications

Landry, M. R. (2025). R/V Roger Revelle Cruise RR2201 BLOOFINZ-IO Cruise Report. Eastern Indian Ocean larval tuna habitat study 20 January – 14 March 2022. <http://hdl.handle.net/1834/43464>  
*General*

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## Parameters

| Parameter        | Description                                                                                                                               | Units           |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Message_ID       | numbered order of message entries                                                                                                         | unitless        |
| Event            | unique event number in UTC time as YYYYMMDD.mmss.###, where ### distinguishes events entered within the same minute                       | unitless        |
| Date             | day of week, day of month, month, year, hh:mm:ss (GMT)                                                                                    | unitless        |
| ISO_DateTime_UTC | Date and time (UTC) in ISO 8601 format                                                                                                    | unitless        |
| Latitude         | latitude (North is positive; South is negative)                                                                                           | decimal degrees |
| Longitude        | longitude (East is positive; West is negative)                                                                                            | decimal degrees |
| Instrument       | name of instrument or sampling system used to collect data                                                                                | unitless        |
| Action           | activity performed with the instrument; e.g. deploy/recover, start/end, etc.                                                              | unitless        |
| Transect_Cycle   | name of sampling transects or transits; lagrangian experiments following a drogued differ are noted as Cycle_Day, e.g. C1_D1, C1_D2, etc. | unitless        |
| Station          | station designations or CTD cast number                                                                                                   | unitless        |
| Cast_Tow         | numbered CTD casts, net tows, sediment traps, drift arrays                                                                                | unitless        |
| Author           | name of person entering the event                                                                                                         | unitless        |
| Comment          | free text comments                                                                                                                        | unitless        |

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## Deployments

### RR2201

|                    |                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Website</b>     | <a href="https://www.bco-dmo.org/deployment/916293">https://www.bco-dmo.org/deployment/916293</a>                                |
| <b>Platform</b>    | R/V Roger Revelle                                                                                                                |
| <b>Report</b>      | <a href="http://hdl.handle.net/1834/43464">http://hdl.handle.net/1834/43464</a>                                                  |
| <b>Start Date</b>  | 2022-01-20                                                                                                                       |
| <b>End Date</b>    | 2022-03-14                                                                                                                       |
| <b>Description</b> | See more information at R2R: <a href="https://www.rvdata.us/search/cruise/RR2201">https://www.rvdata.us/search/cruise/RR2201</a> |

## Project Information

### **Collaborative Research: Mesoscale variability in nitrogen sources and food-web dynamics supporting larval southern bluefin tuna in the eastern Indian Ocean (BLOOFINZ-IO)**

**Coverage:** Eastern Indian Ocean, Indonesian Throughflow area, and the Gulf of Mexico

#### *NSF Award Abstract:*

The small area between NW Australia and Indonesia in the eastern Indian Ocean (IO) is the only known spawning ground of Southern Bluefin Tuna (SBT), a critically endangered top marine predator. Adult SBT migrate thousands of miles each year from high latitude feeding areas to lay their eggs in these tropical waters, where food concentrations on average are below levels that can support optimal feeding and growth of their larvae. Many critical aspects of this habitat are poorly known, such as the main source of nitrogen nutrient that sustains system productivity, how the planktonic food web operates to produce the unusual types of zooplankton prey that tuna larvae prefer, and how environmental differences in habitat quality associated with ocean fronts and eddies might be utilized by adult spawning tuna to give their larvae a greater chance for rapid growth and survival success. This project investigates these questions on a 38-day expedition in early 2021, during the peak time of SBT spawning. This project is a US contribution to the 2nd International Indian Ocean Expedition (IIOE-2) that advances understanding of biogeochemical and ecological dynamics in the poorly studied eastern IO. This is the first detailed study of nitrogen and carbon cycling in the region linking Pacific and IO waters. The shared dietary preferences of SBT larvae with those of other large tuna and billfish species may also make the insights gained broadly applicable to understanding larval recruitment issues for top consumers in other marine ecosystems. New information from the study will enhance international management efforts for SBT. The shared larval dietary preferences of large tuna and billfish species may also extend the insights gained broadly to many other marine top consumers, including Atlantic bluefin tuna that spawn in US waters of the Gulf of Mexico. The end-to-end study approach, highlights connections among physical environmental variability, biogeochemistry, and plankton food webs leading to charismatic and economically valuable fish production, is the theme for developing educational tools and modules through the "scientists-in-the-schools" program of the Center for Ocean-Atmospheric Prediction Studies at Florida State University, through a program for enhancing STEM learning pathways for underrepresented students in Hawaii, and through public outreach products for display at the Birch Aquarium in San Diego. The study also aims to support an immersive field experience to introduce talented high school students to marine research, with the goal of developing a sustainable marine-related educational program for underrepresented students in rural northwestern Florida.

Southern Bluefin Tuna (SBT) migrate long distances from high-latitude feeding grounds to spawn exclusively in a small oligotrophic area of the tropical eastern Indian Ocean (IO) that is rich in mesoscale structures, driven by complex currents and seasonally reversing monsoonal winds. To survive, SBT larvae must feed and grow rapidly under environmental conditions that challenge conventional understanding of food-web structure and functional relationships in poor open-ocean systems. The preferred prey of SBT larvae, cladocerans and Corycaeidae copepods, are poorly studied and have widely different implications for trophic transfer efficiencies to larvae. Differences in nitrogen sources - N fixation vs deep nitrate of Pacific origin - to sustain new production in the region also has implications for conditions that may select for prey types (notably cladocerans) that enhance transfer efficiency and growth rates of SBT larvae. The relative importance of these N sources for the IO ecosystem may affect SBT resiliency to projected increased ocean stratification. This research expedition investigates how mesoscale variability in new production, food-web structure and trophic fluxes affects feeding and growth conditions for SBT larvae. Sampling across mesoscale features tests hypothesized relationships linking variability in SBT larval feeding and prey preferences (gut contents), growth rates (otolith analyses) and trophic positions (TP) to the environmental conditions of waters selected by adult spawners. Trophic Positions of larvae and their prey are determined using Compound-Specific Isotope Analyses of Amino Acids (CSIA-AA). Lagrangian experiments investigate underlying process rates and relationships through measurements of water-column  $^{14}\text{C}$  productivity,  $\text{N}_2$  fixation,  $^{15}\text{NO}_3^-$  uptake and nitrification; community biomass and composition (flow cytometry, pigments, microscopy, in situ imaging, genetic analyses); and trophic fluxes through micro- and mesozooplankton grazing, remineralization and export. Biogeochemical and food web elements of the study are linked by CSIA-AA (N source, TP),  $^{15}\text{N}$ -constrained budgets and modeling. The project elements comprise an end-to-end coupled biogeochemistry-

trophic study as has not been done previously for any pelagic ecosystem.

This award reflects NSF's statutory mission and has been deemed worthy of support through evaluation using the Foundation's intellectual merit and broader impacts review criteria.

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## Program Information

### Second International Indian Ocean Expedition (IIOE-2)

**Website:** <https://web.whoi.edu/iioe2/>

**Coverage:** Indian Ocean

*Description from the [program website](#):*

The Second International Indian Ocean Expedition (IIOE-2) is a major global scientific program which will engage the international scientific community in collaborative oceanographic and atmospheric research from coastal environments to the deep sea over the period 2015-2020, revealing new information on the Indian Ocean (i.e. its currents, its influence upon the climate, its marine ecosystems) which is fundamental for future sustainable development and expansion of the Indian Ocean's blue economy. A large number of scientists from research institutions from around the Indian Ocean and beyond are planning their involvement in IIOE-2 in accordance with the overarching six scientific themes of the program. Already some large collaborative research projects are under development, and it is anticipated that by the time these projects are underway, many more will be in planning or about to commence as the scope and global engagement in IIOE-2 grows.

Focused research on the Indian Ocean has a number of benefits for all nations. The Indian Ocean is complex and drives the region's climate including extreme events (e.g. cyclones, droughts, severe rains, waves and storm surges). It is the source of important socio-economic resources (e.g. fisheries, oil and gas exploration/extraction, eco-tourism, and food and energy security) and is the background and focus of many of the region's human populations around its margins. Research and observations supported through IIOE-2 will result in an improved understanding of the ocean's physical and biological oceanography, and related air-ocean climate interactions (both in the short-term and long-term). The IIOE-2's program will complement and harmonise with other regional programs underway and collectively the outcomes of IIOE-2 will be of huge benefit to individual and regional sustainable development as the information is a critical component of improved decision making in areas such as maritime services and safety, environmental management, climate monitoring and prediction, food and energy security.

IIOE-2 activities will also include a significant focus on building the capacity of all nations around the Indian Ocean to understand and apply observational data or research outputs for their own socio-economic requirements and decisions. IIOE-2 capacity building programs will therefore be focused on the translation of the science and information outputs for societal benefit and training of relevant individuals from surrounding nations in these areas.

A Steering Committee was established to support U.S. participation in IIOE-2. More information is available on their website at <https://web.whoi.edu/iioe2/>.

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## Funding

| <b>Funding Source</b>                                    | <b>Award</b>                |
|----------------------------------------------------------|-----------------------------|
| <a href="#">NSF Division of Ocean Sciences (NSF OCE)</a> | <a href="#">OCE-1851558</a> |
| <a href="#">NSF Division of Ocean Sciences (NSF OCE)</a> | <a href="#">OCE-1851347</a> |
| <a href="#">NSF Division of Ocean Sciences (NSF OCE)</a> | <a href="#">OCE-1851381</a> |
| <a href="#">NSF Division of Ocean Sciences (NSF OCE)</a> | <a href="#">OCE-1851395</a> |
| <a href="#">NSF Division of Ocean Sciences (NSF OCE)</a> | <a href="#">OCE-2404504</a> |

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